Task 3: Multi-Table Operations & Joins

SQL Solutions | MySQL Employees Sample Database

Queries, explanations and illustrative sample outputs

This guide walks through eight multi-table SQL scenarios: joins, set operations, subqueries, aggregation, indexing and common error fixes. Sample outputs are illustrative.

1 INNER JOIN: Employee and Department Mapping

```sql
SELECT
  e.emp_no,
  CONCAT(e.first_name, ' ', e.last_name) AS name,
  d.dept_name
FROM employees e
INNER JOIN dept_emp de ON e.emp_no = de.emp_no
INNER JOIN departments d ON de.dept_no = d.dept_no
WHERE de.to_date > CURDATE()
LIMIT 5;
```

Explanation Displays employee IDs, full names, and department names for current department assignments.

SAMPLE OUTPUT

emp_no	name	dept_name
10001	Georgi Facello	Development
10002	Bezalel Simmel	Sales
10003	Parto Bamford	Production

2 LEFT JOIN: Employees with Manager Information

```sql
SELECT
  e.emp_no,
  CONCAT(e.first_name, ' ', e.last_name) AS employee,
  CONCAT(m.first_name, ' ', m.last_name) AS manager
FROM employees e
LEFT JOIN dept_emp de
  ON e.emp_no = de.emp_no AND de.to_date > CURDATE()
LEFT JOIN dept_manager dm
  ON dm.dept_no = de.dept_no AND dm.to_date > CURDATE()
LEFT JOIN employees m ON dm.emp_no = m.emp_no
LIMIT 5;
```

Explanation Shows employees and their current department managers. Employees without a manager remain in the result with NULL in the manager column.

3 UNION ALL: Active and Former Sales Employees

```sql
SELECT de.emp_no, 'Active' AS status, d.dept_name
FROM dept_emp de
JOIN departments d ON de.dept_no = d.dept_no
WHERE de.to_date > CURDATE() AND d.dept_name = 'Sales'
UNION ALL
SELECT de.emp_no, 'Former' AS status, d.dept_name
FROM dept_emp de
JOIN departments d ON de.dept_no = d.dept_no
WHERE de.to_date <= CURDATE() AND d.dept_name = 'Sales'
LIMIT 5;
```

Explanation Combines current and former Sales department assignments into one result. UNION ALL preserves duplicate rows if present.

SAMPLE OUTPUT

emp_no	status	dept_name
10002	Active	Sales
10005	Former	Sales

4 Correlated Subquery: Current Salary

```sql
SELECT
  e.emp_no,
  e.first_name,
  (SELECT s.salary
   FROM salaries s
   WHERE s.emp_no = e.emp_no
     AND s.to_date > CURDATE()) AS current_salary
FROM employees e
WHERE e.emp_no BETWEEN 10001 AND 10010;
```

Explanation Retrieves each employee's current salary using a subquery linked to the outer employee row.

SAMPLE OUTPUT

emp_no	first_name	current_salary
10001	Georgi	88,977
10002	Bezalel	72,516

5 DISTINCT: Unique Current Job Titles

```sql
SELECT DISTINCT t.title AS unique_titles
FROM titles t
WHERE t.to_date > CURDATE();
```

Explanation Returns unique job titles from records considered current.

SAMPLE OUTPUT

unique_titles
Senior Engineer
Staff
Senior Staff
Manager

6 Complex Scenario: Department Gender Ratios

```sql
SELECT
  d.dept_name,
  ROUND(100.0 * SUM(IF(e.gender = 'F', 1, 0)) / COUNT(*), 1) AS female_pct,
  ROUND(100.0 * SUM(IF(e.gender = 'M', 1, 0)) / COUNT(*), 1) AS male_pct
FROM departments d
JOIN dept_emp de ON d.dept_no = de.dept_no
JOIN employees e ON de.emp_no = e.emp_no
WHERE de.to_date > CURDATE()
GROUP BY d.dept_name
HAVING COUNT(*) > 10000;
```

Explanation Calculates female and male percentages for current department assignments and filters groups with more than 10,000 rows.

SAMPLE OUTPUT

dept_name	female_pct	male_pct
Customer Service	60.2	39.8
Development	42.7	57.3
Sales	31.1	68.9

7 Join Performance Optimization

```sql
CREATE INDEX idx_emp ON salaries (emp_no, salary, to_date);

EXPLAIN ANALYZE
SELECT e.emp_no, MAX(s.salary) AS max_salary
FROM employees e
JOIN salaries s ON e.emp_no = s.emp_no
GROUP BY e.emp_no
LIMIT 10;
```

Explanation The index may improve data access. EXPLAIN ANALYZE reports actual execution details. The source PDF reports an example improvement from 8.2 seconds to 0.9 seconds; actual performance varies.

8 Common SQL Errors and Solutions

```sql
-- Error 1: Avoid an unintended Cartesian product
SELECT *
FROM employees e
JOIN dept_emp de ON e.emp_no = de.emp_no;

-- Error 2: Replace NULL department names
SELECT e.emp_no, COALESCE(d.dept_name, 'Unassigned') AS dept
FROM employees e
LEFT JOIN dept_emp de ON e.emp_no = de.emp_no
LEFT JOIN departments d ON de.dept_no = d.dept_no;

-- Error 3: UNION queries need matching column counts and compatible types
SELECT emp_no AS id FROM employees
UNION
SELECT dept_no AS id FROM departments;
```

Explanation Use explicit join conditions, COALESCE() to replace NULL values, and matching output columns in UNION queries.

* Quick Revision

SQL concept	Purpose
INNER JOIN	Returns rows that match across joined tables.
LEFT JOIN	Keeps every row from the left table.
UNION ALL	Combines results and preserves duplicates.
Correlated subquery	Uses values from the outer query.
DISTINCT	Removes duplicate values.
GROUP BY / HAVING	Groups rows and filters aggregated groups.
COALESCE()	Replaces NULL with a chosen value.
CREATE INDEX	Can improve data retrieval performance.
EXPLAIN ANALYZE	Shows actual query execution details.

Note Sample outputs are illustrative and based on the uploaded PDF; results may vary with database contents and current date.